

Unit IV

4

Polymers in Engineering

Syllabus

Introduction, compounding of plastics, structure, properties and applications of biodegradable polymer, conducting polymer, electroluminescent polymers, thermoplastics polymer and polymer Composites (FRP).

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- 4.1 Introduction
- 4.2 Engineering Thermoplastics
- 4.3 Biodegradable Polymers
- 4.4 Conducting Polymers
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- 4.7 Compounding of Plastics

Course Objective

To understand structure, properties and applications of speciality polymers.

Course Outcome (CO4)

Demonstrate the knowledge of advanced engineering materials for various engineering applications.

4.1 Introduction

- The commercial application of polymers have increased over the years. Wide variety of polymers are used for engineering and medical applications. The demand for polymers with specific properties for the specific end use has increased. This has lead to the development of new polymeric materials speciality polymers.
- Speciality polymers exhibit exclusively remarkable performance in certain special, limited but critical applications such as in aerospace composites, membrane for gas liquid separations, fire retardant textile fabric used by fire fighters, sutures and surgical implants etc.
- Engineering materials are the substances which are useful in the field of engineering. Engineering materials refer to selecting the correct materials for the engineering applications. Engineers will choose a particular grade or type of material based on its properties. A particular material is selected on the basis of following concerns.
 - **Physical properties** : Density, specific heat, thermal expansion etc.
 - **Chemical properties** : Oxidation, corrosion, flammability, toxicity etc.
 - **Mechanical properties** : Strength, ductility, toughness etc.
 - **Manufacturing properties** : Formed, casting machined, welding, etc.

Classification of engineering materials :

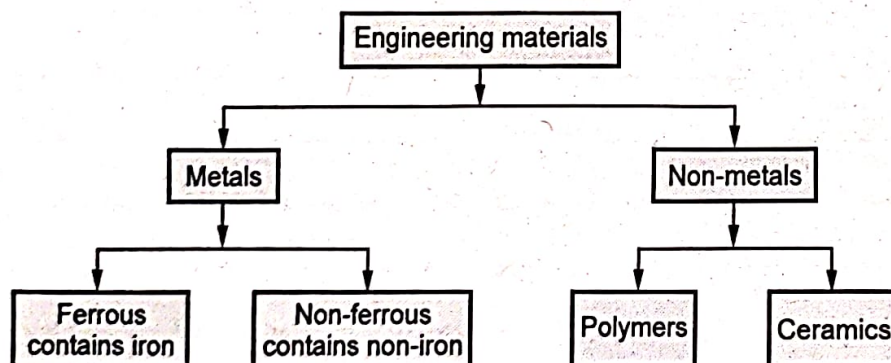
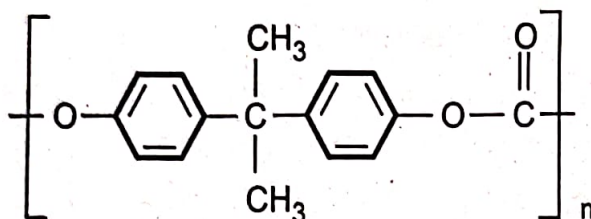


Fig. 4.1.1 Speciality polymers

- Speciality polymers possess magnificent (Excellent) properties in one or more are as when compared with commodity polymers. Superiority of specialty polymers may include excellent chemical resistance or high thermal stability, light weight, low creep compliance and high tensile strength. These polymers can be the replacement for metals, usually where as high strength to weight ratio is an important parameter e.g. in automotive and aerospace applications.

4.2 Engineering Thermoplastics

- Engineering thermoplastics are a group of plastic materials that have better mechanical and / or thermal properties than the more widely used commodity plastics (Such as polystyrene, PVC, polypropylene and polyethylene). Engineering thermoplastic is obtained from high polymer resins, which possess :
 - i) High load-bearing characteristics.
 - ii) Fairly good thermal stability.
 - iii) High mechanical strength.
 - iv) Readily mouldable characteristics into complicated shapes.
 - v) High abrasion resistance.
 - vi) High dimensional stability.
 - vii) Light weight.
 - viii) High dielectric constant.
 - ix) Rigidity.
 - x) High performance properties.
- These properties permit them to be used in the same manner as metals, alloys and ceramics.
- Engineering thermoplastics not only replace metals, but they can also be used along with metals. They meet the crucial needs of the day with greater efficiency and reduction in both cost and weight. They are finding applications in demanding areas like automobiles, defence, electrical and electronics, telecommunications, satellite robots, computer components etc.
- Examples of engineering plastics (polymers) include polycarbonate, Acrylonitrile Butadiene Styrene (ABS), polyamides (nylons), polysulphone (PSU), polytetrafluoroethylene (PTFE / Telfon) etc.

4.2.1 Polycarbonate (Lexan, Merlon)**Structure****Properties :**

- i) It has a very high impact strength and tensile strength over wide range of temperature.
- ii) It is a highly transparent plastic. It has higher light transmittance (about 88 %) than many types of glasses.
- iii) It is resistant to water and many organic compounds but are soluble in number of organic solvents and alkalis.
- iv) It has a good heat resistance (Useful upto 140 °C).
- v) It has low combustibility. It has high melting point. It melts around 265 °C.

Applications :

Polycarbonate polymers are used to produce a variety of products.

- i) It can be used in CDs, DVDs and blue-ray discs.
- ii) It is used as bullet-proof material vests, due to its impact strength and transparency.
- iii) It is used as insulators in electrical and telecommunication hardware.
- iv) It is used for making sunglasses, swimming and SCUBA goggles, safety goggles, helmets visor, covers of vehicle light.
- v) It is used for machinery housings.
- vi) It is a suitable alternative to glass in a number of glazing applications such as agricultural houses, security windows, industry or public building.
- vii) PC and its blend are used in appliances such as refrigerators, air conditioner, coffee machines, food mixers etc. because it offers shape freedom through wide range of mechanical properties.
- viii) Food storage containers made from PC are reusable and can be conveniently used in refrigerator and microwave.

Review Questions

1. What is engineering thermoplastic ? Give the structure of polycarbonate and its applications.
2. State the structure, properties and applications of polycarbonate.

4.3 Biodegradable Polymers

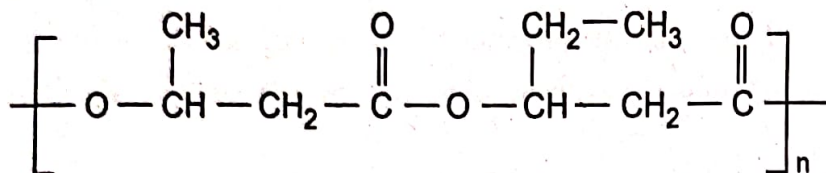
- Biodegradable polymers are the polymers in which degradation results from the action of naturally occurring micro-organisms such as bacteria, fungi and algae via enzymatic activity.
- The process of converting or degrading polymers into harmless simple products via enzymatic action of micro-organism is called **biodegradation**.

4.3.1 Factors affecting Biodegradation

- i) Micro-organism ii) Environment iii) Nature of polymer.
- Micro-organisms** : Micro-organisms like bacteria, fungi and algae play key role for biodegradation of polymers. They break C-C bond by aerobic or anaerobic way, thereby degrades polymers into harmless simple products. The rate of this process is very slow.
 - Environment** : The favorable environment is very important for survival, multiplication and action of micro-organisms, which include suitable temperature, moisture, presence of salts, oxygen, pH etc.
 - Nature of polymers** : Biodegradation process takes place by hydrolysis and oxidation, so the polymer chain should possess bonds which can be easily hydrolyzed or oxidized by the enzymatic action of micro-organisms. There should be N, O, S like atoms in the chain of polymers.

4.3.2 Classification of Biodegradable Polymers

- Biodegradable polymers can be broadly classified as,
 - Natural biodegradable polymers.
 - Synthetic biodegradable polymers.
 - Biosynthetic biodegradable polymers.
- i) Natural biodegradable polymers** : Natural biodegradable polymers are produced in nature from plant or animal resources. e.g. starch, cellulose, protein etc.
- ii) Synthetic biodegradable polymers** : Synthetic biodegradable polymers are produced by mankind. The major category of these polymers consists of aliphatic polyesters with hydrolysable linkage. e.g. polylactic acid, polycaprolactone.
- iii) Biosynthetic biopolymers** : Produced by fermentation process by micro-organisms. For example polyhydroxy alkanotes (Polyhydroxy butyrate, polyhydroxy valerate).

4.3.3 Poly (Hydroxybutyrate - Hydroxy Valarate)

PHBV or PHBHV

Properties :

- i) It is biodegradable, soft and flexible.
- ii) Easily mouldable, can be converted to films, thermosoftening polymer.
- iii) It has oil resistance and moisture resistance.
- iv) It is easily hydrolysed above pH₉ and below pH₃.
- v) On aerobic biodegradation final product is CO₂ and water, while during anaerobic condition final product is methane and CO₂.

Applications :

- i) **Packaging** : Biopol can be converted to films which are useful for packaging, lamination, carry bags, disposable bottles.
- ii) **Moulding articles** : Can be used for injection moulding, blow moulding, extrusion moulding, articles of common consumer applications.
- iii) **Medical field** : It is important in biomedical applications like surgical sutures, organs regeneration, orthopedic treatment, slow release drugs.
- iv) **Agriculture** : Biodegradable polymers are preferred in agriculture for mulching, netting, twine etc. for crop yields and reducing crop management cost. They are also required for controlled release of fertilizer, pesticide.

Review Questions

1. Define biodegradation of polymers. State favourable structure of polymer for biodegradation. Write structure of biopol (PHBV).
2. What is biodegradable polymer ? Give the structure of PHBV and its applications.
3. What is biodegradable polymer ? Draw the structure of PHBV and write its applications.

4.4 Conducting Polymers

- Organic polymers in general, were used in electronic, electrical and other applications as insulators due to their high resistance to the flow of current.
- Now, organic polymers having electrical conductance of the order of conductors are called **conducting polymers**. These can be classified as extrinsically and intrinsically conducting polymers.

- **Extrinsically conducting polymers** : Extrinsically conducting polymers are prepared by mixing conducting fillers like metal fibers, metal oxides, carbon black with insulating polymers to make it conducting.
- **Intrinsically conducting polymers** : Intrinsically conducting polymers shows conductivity due to presence of conjugation and planarity in a molecule. Due to which electrons movement takes place, causes the conductivity in a polymer. e.g. Trans-Polyacetylene, Poly (p-phenylene), Polypyrrole, etc.
- They conduct electricity when doped with oxidising or reducing agents or protonic acids.
- The basic requirement for organic polymers to be inherently conducting is the formation of continuous conjugation (Alternative single and double bonds). The following are examples of intrinsically conducting polymers.

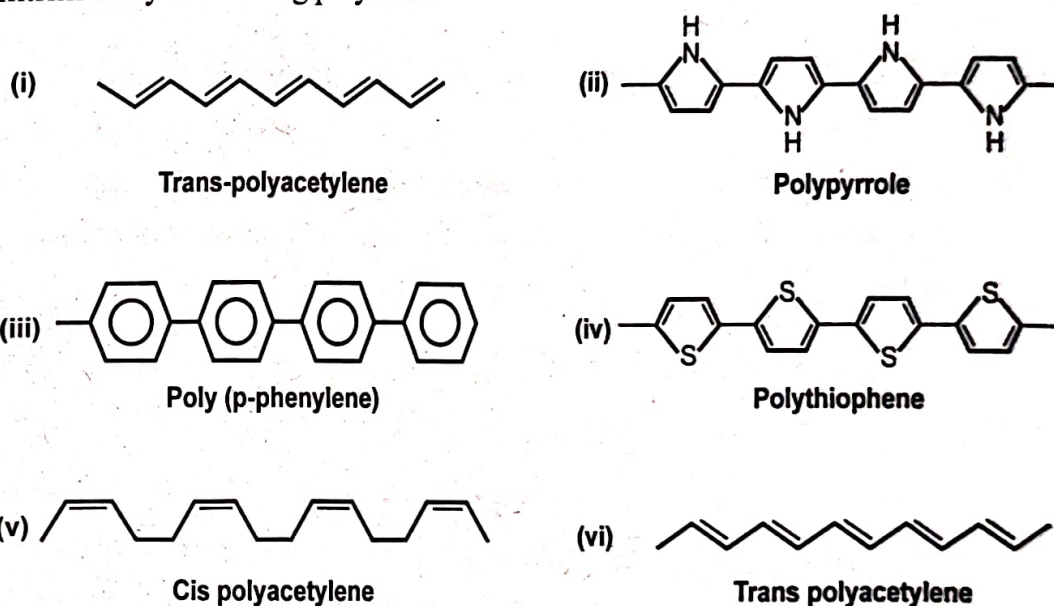


Fig. 4.4.1

4.4.1 Polyacetylene

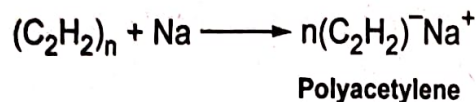
- Polyacetylene consists of a long chain carbon atoms with alternating single and double bonds.

Doping of polymers :

- Prof. Alan MacDiarmid, Prof. Alan Heeger and Prof. Hideki Shirakawa found that the conductivity of polymer can be increased to the extent of metals by doping. Polyacetylene in its original form is semiconductor. Doped polyacetylene has good conductivity such as copper and silver.
- **Types of doping :**
 - ◆ **p-doping (Oxidation)** : p-doping is carried out through oxidation.



♦ **n-doping (Reduction)** : n-doping is carried out through reduction.



Conduction mechanism in polyacetylene :

- Conductivity of pure acetylene is about 4.4×10^{-5} S/cm. Upon doping with oxidizing agents like iodine, the conductivity increases to about 400 S/cm.
- i) When the oxidative dopant such as iodine is added to it takes away an electron from the - backbone of the polyacetylene chain and creates a positive centre (hole) on one of the carbon.
- ii) The other - electron resides on the other carbon making it a radical. The radical ion formed is called **polaron**.
- iii) A dipolaron (Soliton) is formed on further oxidation. Then these radicals migrate and combine to establish a backbone double bond.
- iv) As the two electrons are removed, the chain will have two positive centres (Holes). When a potential is applied, they migrate from one carbon to another and account for conductivity.

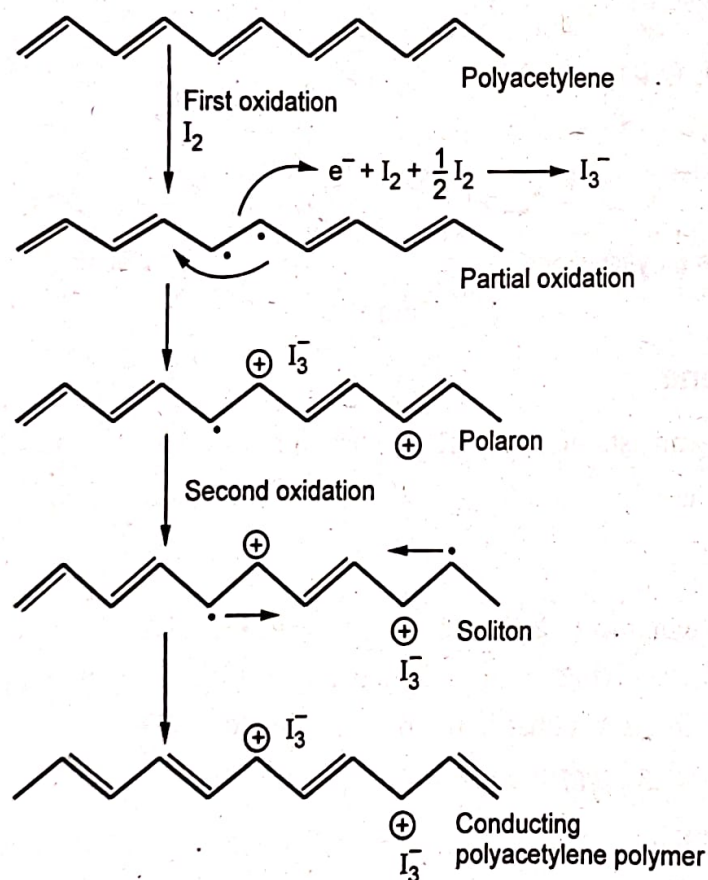


Fig. 4.4.2

- Before doping there is sufficient energy gap between VB and CB, hence electrons remain in VB and the polymer acts as an insulator.
- Upon doping, when sufficient solitons are formed, a new mid-gap energy band is created which overlaps with the valence and conduction bands allowing electrons to flow. The charged solitons are thus responsible for making polyacetylene a conductor.

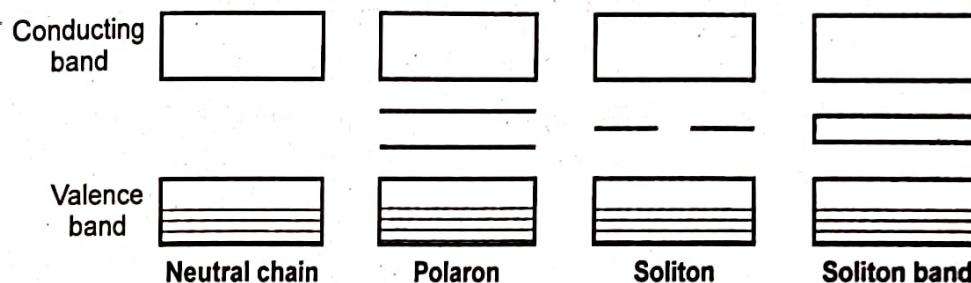


Fig. 4.4.3 Formation of new mid-gap energy band between VB and CB

Properties of polyacetylene :

- Films of cis-polyacetylene are very flexible and can be readily stretched, while trans-poly-acetylene is much more brittle.
- Polyacetylene has a bulk density of 0.4 g/cm^3 , while density of the foam is significantly lower, $0.02 - 0.04 \text{ g/cm}^3$.
- Exposure to air causes a large decrease in the flexibility and conductivity.

Applications :

- Doped poly-acetylene have a particular high electrical conductivity, so it is used as promising charge storage material.
- It is used as a sensor to measure glucose concentration.
- It can be used as antistatic material. Coating an insulator with conductive polymer is also important for explosives industry, computer industry as antistatic materials.

Review Questions

- What are intrinsic and extrinsic polymers? Explain with their examples.
- What is conducting polymer? Give the structural requirements and its applications.

4.5 Electroluminescent Polymers

- Electroluminescence is an optical and electrical phenomenon in which a material emits light in response to the passage of an electric current or to a strong electric field.
- Electroluminescent polymers are organic polymers that emit light upon application of an electric field.

- Light emission in the process of electroluminescence occurs by recombination of charge carriers of opposite sign (electrons and holes) injected into the polymer thin film through electrodes (Cathode and anode, respectively) from an external circuit. The resulting device is a Polymer Light-Emitting Diode (PIED) or, more in general, an Organic Light-Emitting Diode (OLED).
- OLEDs have attracted great technological interest for fabrication of flat panel displays (e.g. TV and computer screens) and for low energy consumption.
- A common structural pattern in the electroluminescent polymers is that throughout the hole polymeric backbone chain consisting of alternating single and double or triple C–C bonds.
- A common structural pattern in the electroluminescent polymers is that, throughout the whole polymeric backbone chain consisting of alternating single and double bond or triple C–C bonds (vinylene, ethylene, aromatic, heteroaromatic units and their combination).

4.5.1 Construction and Working of OLED

- In the simplest setup of an organic electroluminescent light emitting diode, a thin layer of electroluminescent polymer is sandwiched between two electrodes (The anode and the cathode), one of which at least, must be transparent. Generally, cathode consists of metal layer such as Al or Ca, whereas Indium Tin Oxide (ITO) is commonly used as transparent anode supported on a glass or plastic substrate.

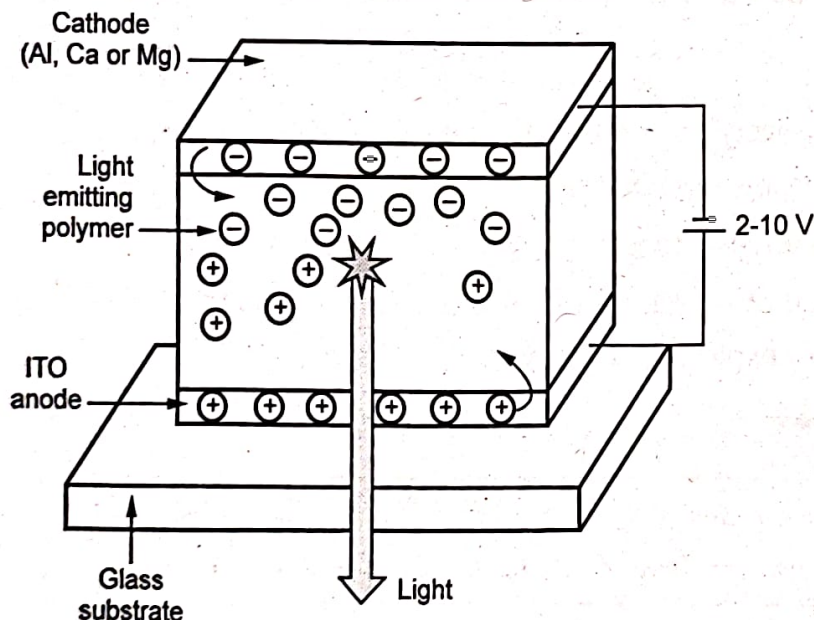
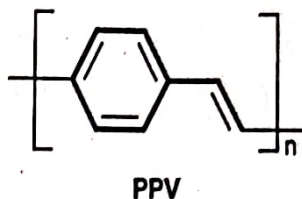


Fig. 4.5.1 Schematic drawing of polymer based OLED

- Upon the application of voltage (2-10 V), electron and holes are emitted into the emitting polymer layer from the cathode and anode respectively. Recombination of an electron-hole pair within the polymer may result in the emission of light (Photon).

4.5.2 Polyphenylene Vinylene

- PPV is an electroluminescent material.



Properties :

- PPV is a diamagnetic material.
- It has a very low intrinsic electrical conductivity of the order of 10-13 S/cm.
- The electrical conductivity increases upon doping with iodine, ferric chloride, alkali metals or acids. Though, the stability of these doped material is relatively low.
- Alkoxy substituted PPVs are generally easily oxidized than the parent PPV and hence have much higher conductivities.
- It shows bright yellow-green fluorescence on application of electric field.

Applications :

- As PPV is electroluminescent material, it shows electroluminescence, leading to applications in polymer base OLEDs.
- It is also used as an electron donating material in organic solar cells.
- PPV is used for fabrication of flat panel display devices (TV and computer screen).
- It is also used in automotive instrument panel backlighting.
- It is used in photovoltaic cells.

Review Question

1. Give structure, properties and applications of poly-phenylenevinylene (PPV).

4.6 Polymer Composites

- Current technology required such structural materials, should have the properties such as low density, high strength and stiffness, good abrasion and impact resistance and good corrosion resistance. Any single metal, alloy, ceramic or polymeric material cannot offer the combination of the above properties. The search for materials possessing properties required by modern technology has led to the development of composite materials.

- A composite is a combined material formed by the assembly of two or more components, such as fillers or reinforcing agents and a compatible matrix binder in the order to obtain specific characteristic properties.
- Two or more distinct components which combine to form a new class of material suitable for structural application are referred as composite materials.

Constituents of composites :

Composite materials have two major phases.

- Matrix phase :** It is the primary phase having a continuous character and gives a bulk form to the composite. It holds the dispersed phase and share a load with it. Polymer matrix includes thermoplastic material / (acetyl, ABS, PE, PP) and thermoset resin.
- Dispersed/Reinforcing phase :** This phase is embedded in the matrix in a discontinuous form and provide high strength, rigidity and enhance matrix properties.

Classification of composites :

In the basis of reinforcement form, composites can be classified as :

- Fiber reinforced composites :** These are composed of fibers embedded in the matrix material. For example, glass, carbon, aramides, fibres etc.
- Laminar composites :** These are composed of layers of materials held together by matrix. Sandwich structures fall under this category.
- Particulate composites :** These are composed of particles distributed or embedded in a matrix body. The particles may be flakes or in powder.

4.6.1 Fiber Reinforced Plastic (FRP)

- Fiber reinforced plastics are composites composed of fibers and a polymer matrix. Fibers impart reinforcement and are the main source of strength, while the polymer matrix glues all the fibers together in shape and transfers stresses between the reinforcing fibers.

Processing of fiber reinforced plastics :

- There are many techniques available for processing of FRPs.
- In filament winding, the fiber is wound to form a desired pattern usually but not necessarily hollow and cylindrical. The fiber is passed through the resin and then spun onto a mandrel. After desired number of layers of fiber is added, it is cured. With the advent of new machinery, complex shapes and designs of the filament can be readily wound.
- Some important and useful properties of FRP are listed as follows :
 - i) They are light in weight as compared to traditional materials.

- ii) They have high strength per unit of weight.
- iii) They are highly resistant to chemicals.
- iv) They offer greater design flexibility.
- v) They are also heat resistant.

Some important types of fibre reinforced composites are as

- 1) Glass fiber reinforced polymer
- 2) Carbon reinforced polymer composite

4.6.2 Glass Fiber Reinforced Polymer Composite

- Glass fibers are reinforced in polymer matrix containing nylon, polyesters etc.
- Glass fiber geometry in the polymer matrix can be listed as
 - i) **Aligned fibers** : In this type of alignment, the fibres are woven parallel to each other.
 - ii) **Random fibers** : In this type of alignment, the fibres are woven randomly according to its specific density.
 - iii) **Woven fibres** : In this type of alignment, the fibres are woven in the shape of a net. It is strongest among the three.
- Glass fibres are obtained by forcing glass through spinner at (Having small holes) and rapidly pulling and cooling to get fibres.

Properties :

- They have following properties :
 - i) High strength
 - ii) Light weight
 - iii) Resistance
 - iv) Design flexibility
 - v) Seamless construction.

Applications :

- i) Due to their light weight, these composites are used in making cars and aircrafts.
- ii) Due to their strength, composites like graphite epoxy are used in making bridges.
- iii) Used for thermal insulation and sound absorption.
- iv) Because of their high strength and insulation, these composites are used in making armors.

4.6.3 Carbon Reinforced Polymer Composite

- Carbon fiber is reinforced in polymer matrix like epoxy or polyesterresins.
- Carbon fibers is produced from a polymer such as polyacrylonitrile, rayon or petroleum pitch, known as **precursor** in an inert atmosphere.

Properties :

- i) CFRPEs have excellent resistance to corrosion.
- ii) High tensile strength.
- iii) High Young's modulus (i.e. high stiffness or resistance to extension under load).
- iv) High thermal conductivity.
- v) Low density.

Applications :

- i) The strongest demand from aircraft and aerospace, as well as the automotive industry.
- ii) Used in making sports goods.
- iii) CFRP has become a notable material in structural engineering.
- iv) Used for fabrication of carbon fibre micro-electrodes.
- v) Used in laptop shells and other high performance cases.
- vi) In light weight poles such as tripod legs, tent poles, fishing rods, etc.

Review Questions

1. Explain FRP with respect to their properties and applications.
2. What are polymer composites ? Write properties and applications of glass FRP.

4.7 Compounding of Plastics

- A polymeric material alone (Virgin polymer), is not used for making finished final articles.
- It is mixed with few compounding constituents which have some useful functions or impart useful properties to the final product.
- The compounding or mixing of plastics can be with following constituents :

1. Resin or binder :

- This is the thermosetting polymer forming constituent. It is taken in a mould and mixed up with some other constituents, along with catalyst to get the article of final shape. Sometimes a lower molecular weight linear polymer is taken in the mould and mixed with cross linking agent. On heating, it produces a thermoset article.

2. Plasticizer :

- This constituent is added to thermosoftening plastic in small percentage. Commonly used plasticizers are, vegetable oil, ester of phthalic or stearic acid, tributyl phosphate, triphenyl phosphate etc.
- The plasticizer molecules occupy positions between the linear or branched polymer molecules and separate them a little.
- They decrease the intermolecular force of attraction between polymer molecules and cause greater freedom of sliding or movement on heating or on application of stress.
 - i) It reduces softening temperature of plastic and makes moulding easier.
 - ii) It increases plasticity, flexibility of plastic.
 - iii) It reduces chemical and solvent resistance of plastic.
 - iv) It makes polymer more amorphous, decreases strength.
 - v) It reduces T_g of the plastic.

3. Fillers :

- Various cheap materials like saw dust, sand powder, BaSO₄, glass fibres, clay, carbon black, CaCO₃, metal powder, quartz etc. are compounded with thermosoftening plastic. Air is a filler in foamed plastics.
- Functions of fillers are,
 - i) To reduce cost of plastic.
 - ii) To increase mechanical strength of plastic.
 - iii) To impart some special properties to plastics.

Example : Barium salts make plastic X-rays absorbing, glass fibres increase tensile strength of plastic to high extent.

4. Colour or pigments :

- Initially plastic are white / translucent / dull to look.
- To make them look better, some coloured substances are mixed with both thermosoftening and thermosetting plastics.
- **Example :** Organic dyes, carbon black, inorganic oxides of different colours.

5. Lubricants :

- Lubricating oils, waxes, vaselines, soaps like materials are applied to innerside of a mould before a plastic is moulded.
- This helps in easier moulding and glossy finish to moulded article.

- The lubricant prevents plastic material from sticking to moulding equipment.

6. Stabilizers :

- Some thermosoftening plastics decompose and discolour at the moulding temperature.
- Therefore thermal stabilizers like lead salts or cadmium or barium salts in small proportion, are compounded with plastic.

7. Accelerators :

- To make the formation of thermosetting plastic formation during moulding faster, some catalysts like benzoyl peroxide, transition metal oxides, cu, lead, are mixed in very small amounts in the resin. Other compounding constituents like flame retarders, blowing agents, nucleating agents are also mixed in plastics, if needed.

Unit IV - Question Bank

- Q.1** Define compounding of plastics and explain the role of any two additives with *example*.
(**Listing additives - 2 M, Explanation with example - 2 M each**)
- Q.2** Describe the effect of plasticizers and stabilizers in the compounding of plastics with their importance. (**Role of plasticizers - 2 M, Role of stabilizers - 2 M, Importance - 1 M**)
- Q.3** Discuss which ingredients improves the mechanical and thermal properties of plastics by compounding of plastics with examples. (**Mechanical properties - 2 M, Thermal Properties - 2 M, Example - 1 M**)
- Q.4** Define thermoplastics. Draw the structure of polycarbonate and write four physical properties and two applications. (**Definition - 1 M, Structure - 1 M, Properties - 2 M, Applications - 1 M**)
- Q.5** Define thermoplastics. Write four physical properties and four applications.
(**Definition - 1 M, Properties - 2 M, Applications - 2 M**)
- Q.6** Draw the structure of polycarbonate and write four physical properties and two applications.
(**Structure - 1 M, Properties - 2 M, Applications - 2 M**)
- Q.7** Define Biodegradable polymers. Discuss the factors influencing the degradation of polymers along with structure of PHBV. (**Definition - 1 M, Factor - 1 M each, Structure - 1 M**)
- Q.8** Define biodegradable polymers and biodegradation process. Discuss the factors influencing the degradation of polymers. (**Definition - 1 M each, Factor - 1 M each**)
- Q.9** What are biodegradable polymers ? Draw the structure of PHBV with their four properties and two applications. (**Definition - 1 M, Structure - 1 M, Properties - 2 M, Applications - 1 M**)
- Q.10** Define electroluminescent polymers with structure of PPV. Outline the construction and working of an Polymer Light-Emitting Diode (PLED) with neat labelled diagram.
(**Definition - 1 M, Structure - 1 M, Construction - 1 M, Working - 1 M, Diagram - 1 M**)
- Q.11** Draw the structure of PPV. Explain the four properties and four applications of PPV.
(**Structure - 1 M, Properties - 2 M, Applications - 2 M**)
- Q.12** Define conducting polymers with structure of two examples. Explain the structural requirements for a polymer to exhibit electrical conductivity. (**Definition - 1 M, Structure - 1 M each, Requirement - 2 M**)

- Q.13** Classify conducting polymers. Explain the difference between intrinsically conducting polymers and extrinsically conducting polymers with examples. (**Classification - 1 M, Difference - 1 M each**)
- Q.14** What are doped conducting polymers? Compare P-doping and N-doping. (**Definition - 1 M, Difference - 1 M each**)
- Q.15** Explain how doping improves conductivity of polymers with examples of P-doping and N-doping. (**Explanation - 1 M, P-doping - 2 M, N-doping - 2 M**)
- Q.16** Explain composite polymers with classification and their examples. (**Explanation - 1 M, Classification with example - 2 M each**)

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